

Lecture 04 — Image Pyramids

Lecture 4

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Learning Objectives

By the end of this lecture, you should be able to:

- Explain why naïve downsampling can create **aliasing**.
- Describe how **smoothing before downsampling** reduces aliasing.
- Construct and interpret a **Gaussian pyramid**.
- Construct and interpret a **Laplacian pyramid**.
- Explain why Laplacian pyramids are useful for **compression and reconstruction**.
- Recognize how multi-scale representations appear in applications such as blending, tracking, detail enhancement, steerable pyramids, and wavelets.



This image is too big to fit on the screen!
How would you reduce it to half its size?

Naïve Image Downsampling

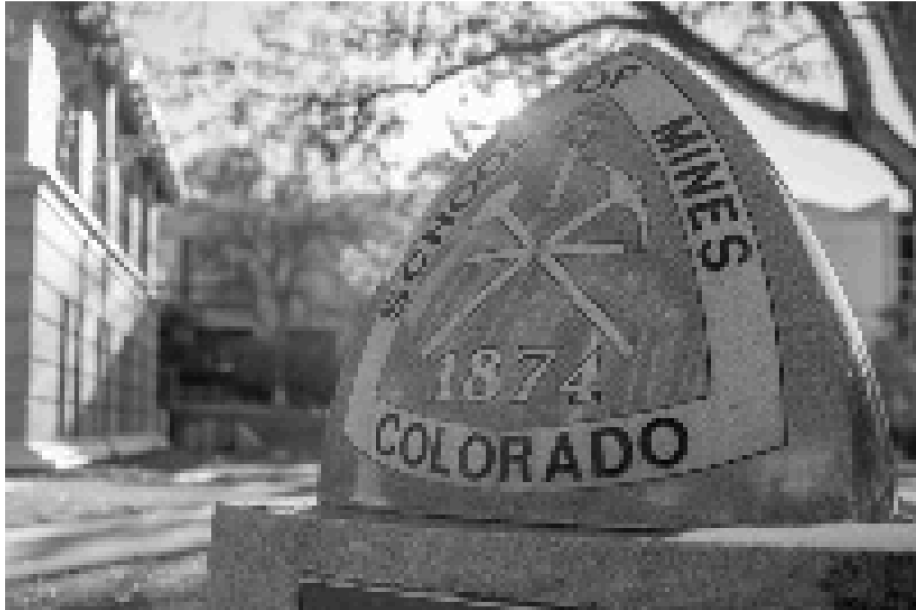
Keep every other row and column, then repeat.

```
1 import matplotlib.pyplot as plt
2
3 y2 = I[:, ::2, ::2]
4 y4 = y2[:, ::2, ::2]
5 y8 = y4[:, ::2, ::2]
6
7 fig, ax = plt.subplots(1, 3, figsize=(8, 4),
8     gridspec_kw={"width_ratios": [4, 2, 1]})
9
10 for a, y, s in zip(ax, [y2, y4, y8], [2, 4, 8]):
11     a.imshow(y, cmap="gray", vmin=0, vmax=255,
12         interpolation="nearest")
13     a.set_title(f"1/{s}")
14     a.axis("off")
15 plt.tight_layout()
16 plt.show()
```



Naïve Image Downsampling

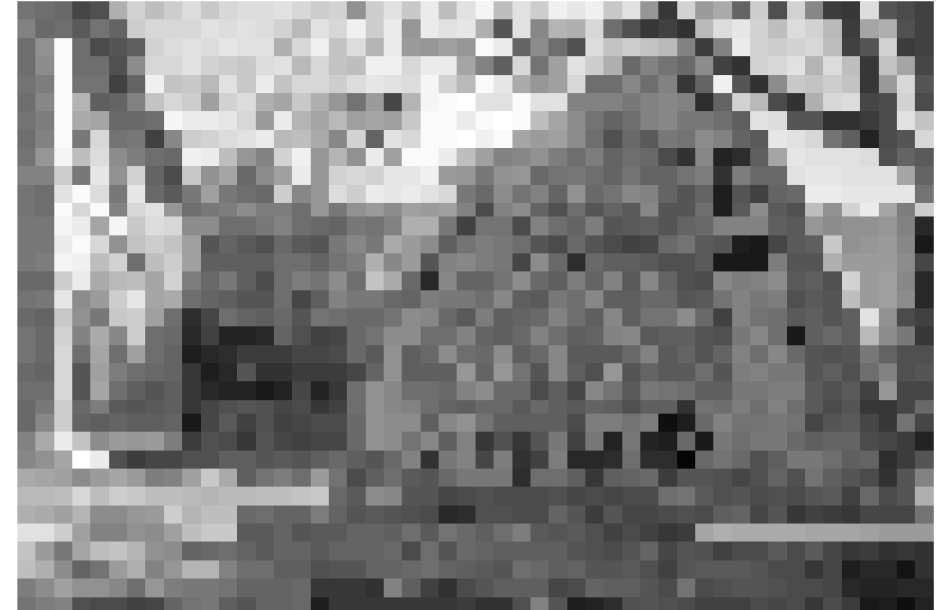
1/2



1/4 (2× zoom)

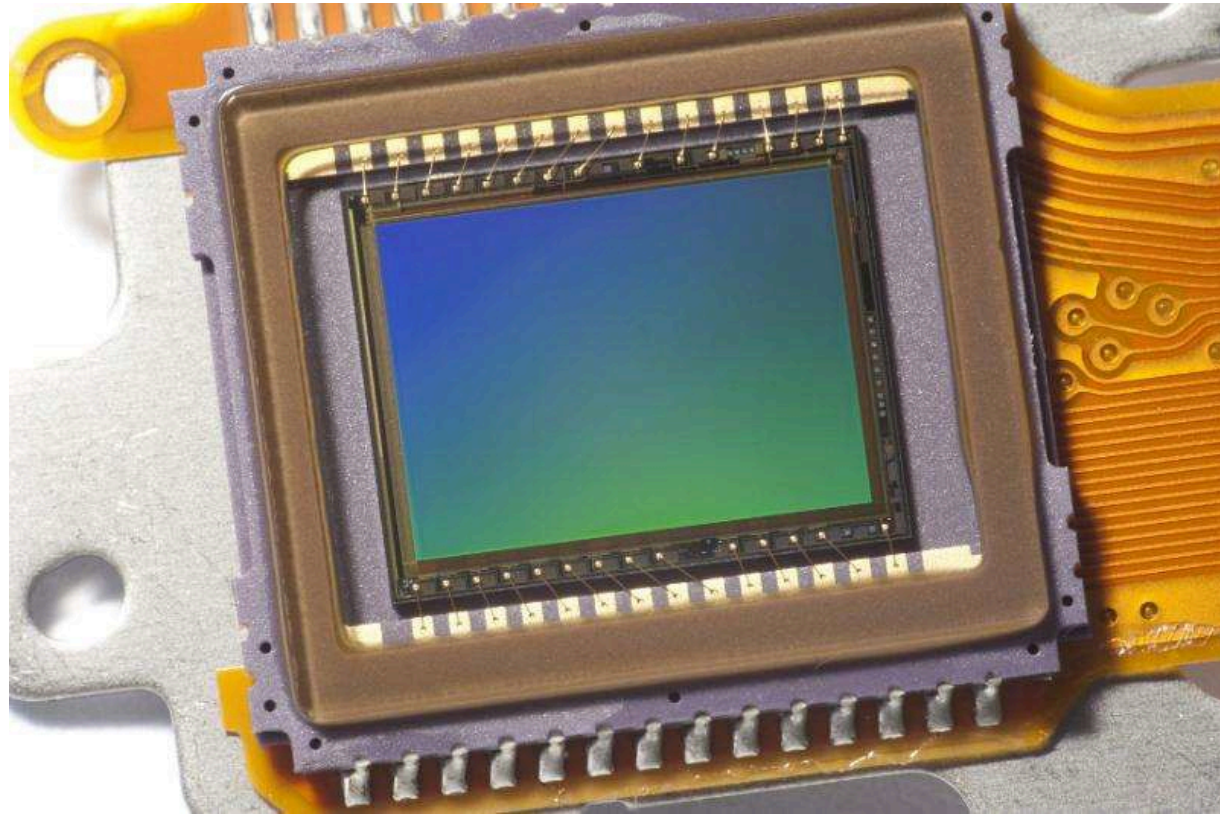


1/8 (4× zoom)

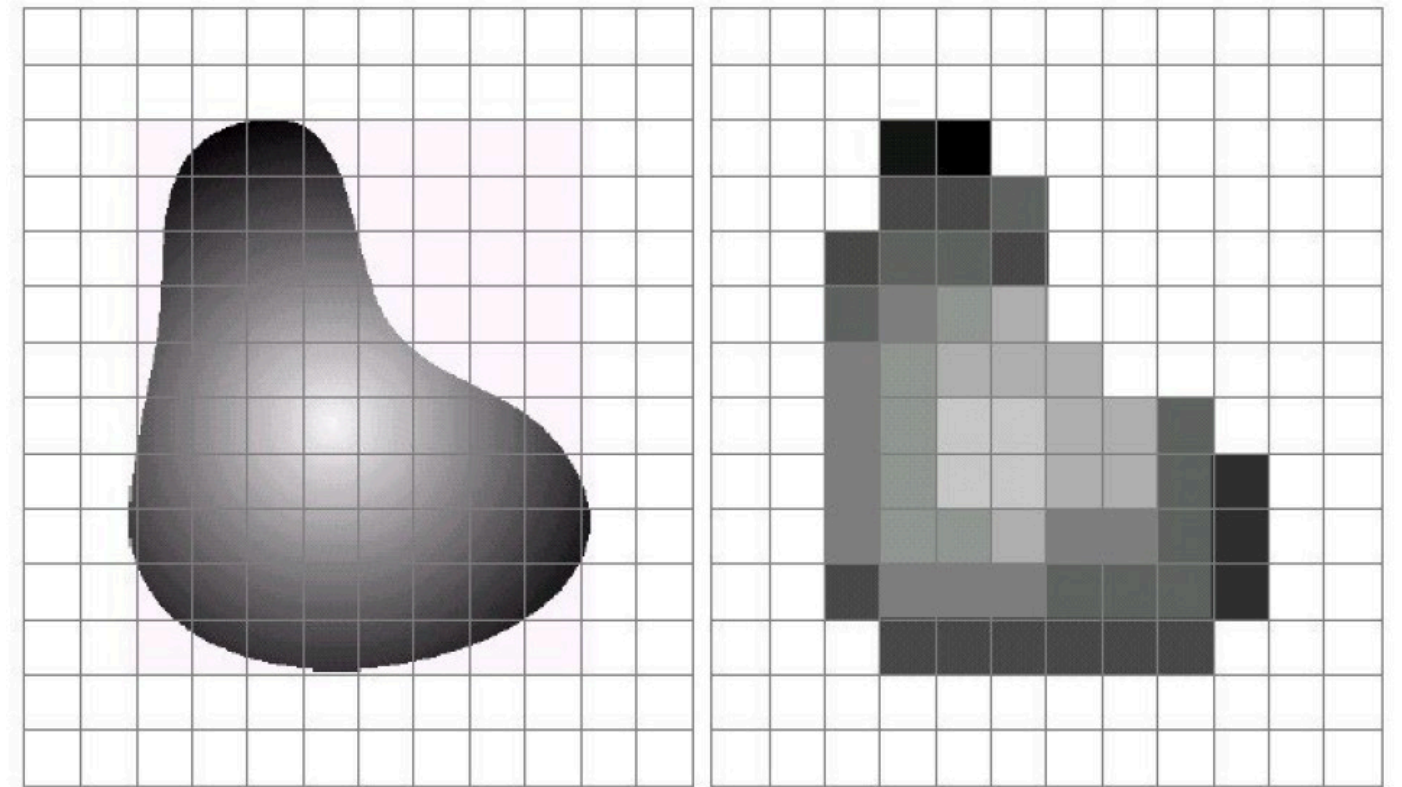


Why is the 1/8 image so pixelated (and do you know what this effect is called)?

Reminder



Camera / sensor hardware

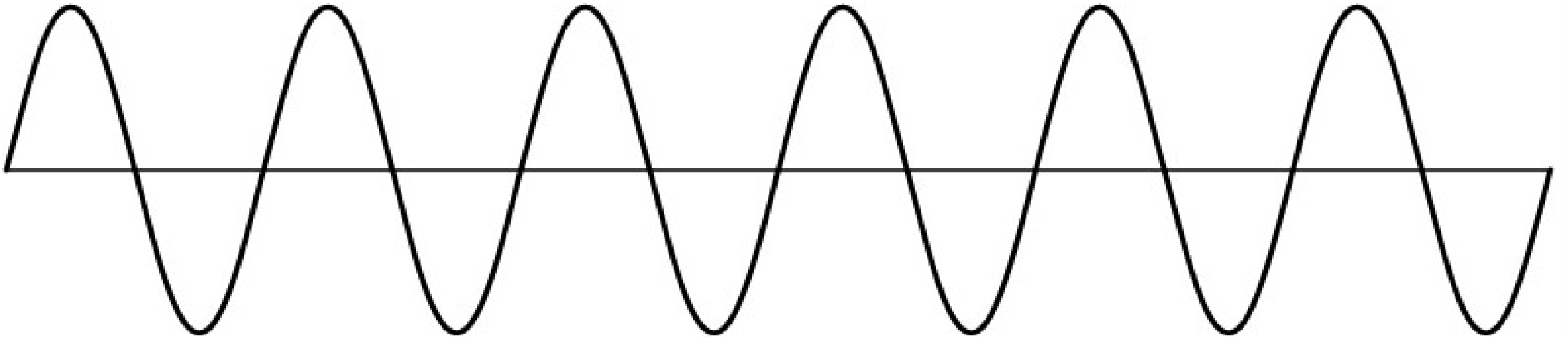


CMOS sensor array

Images are a **discrete**, or **sampled**, representation of a **continuous** world.

Sampling

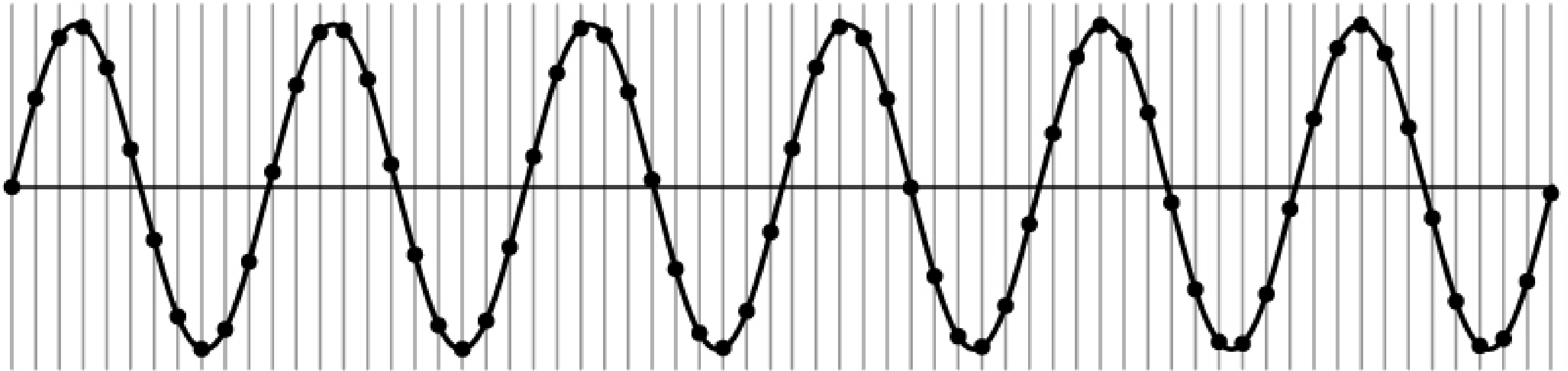
A simple example: a sine wave



How would you discretize this signal?

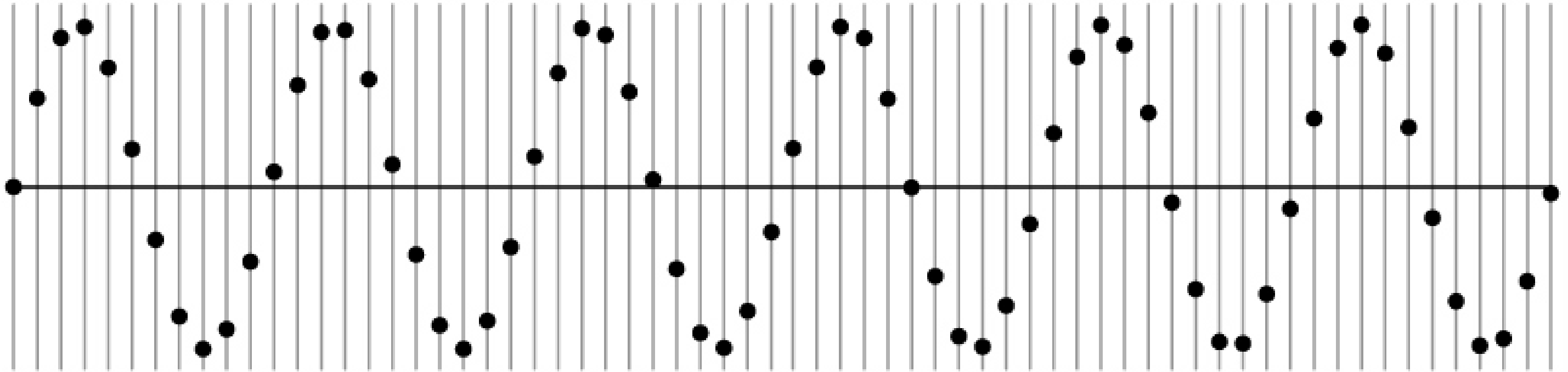
Sampling

Sample the signal at regularly spaced locations.



Sampling

The sampled signal consists of discrete values.

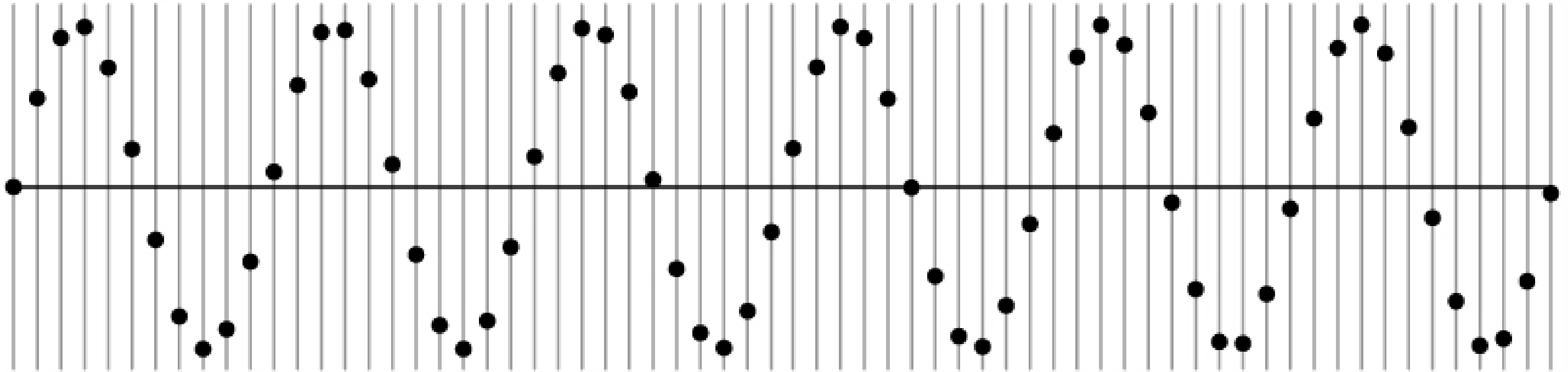


How many samples should I take?

Can I take as **many** samples as I want?

Sampling

The sampled signal consists of discrete values.

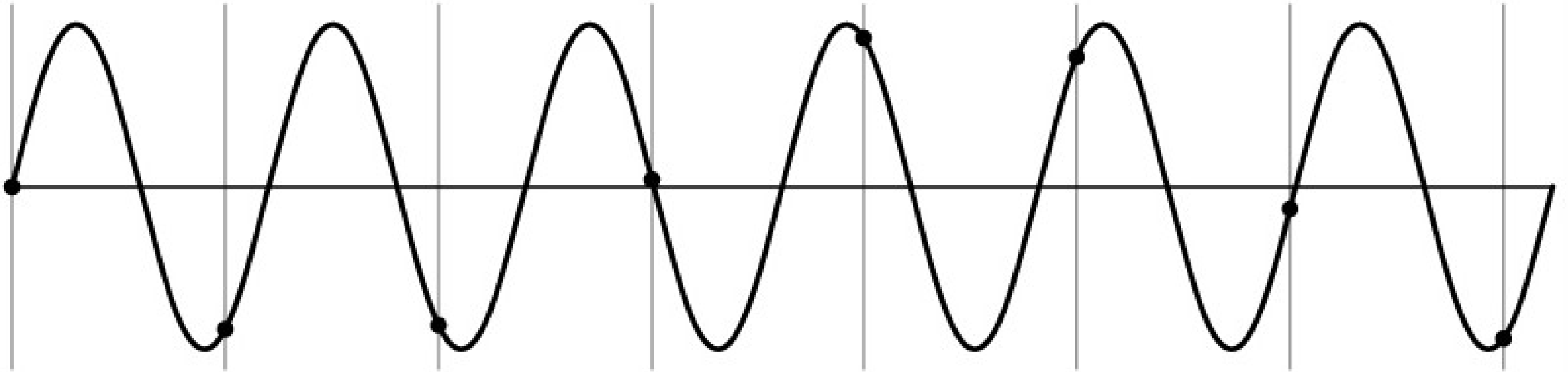


How many samples should I take?

Can I take as **few** samples as I want?

Undersampling

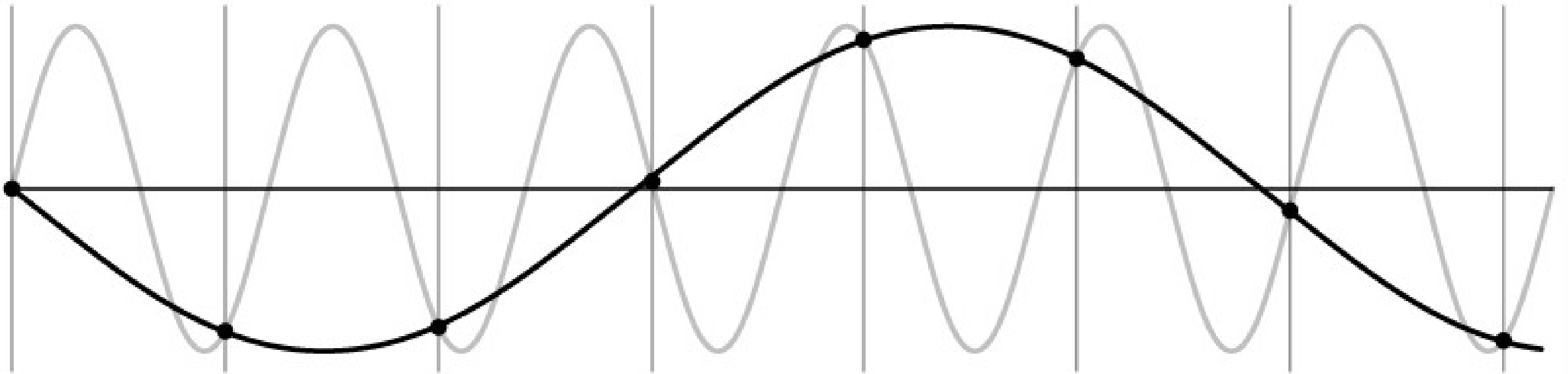
What happens when we take fewer samples?



Unsurprising effect: **information is lost.**

Undersampling

A different signal can agree with the same samples.

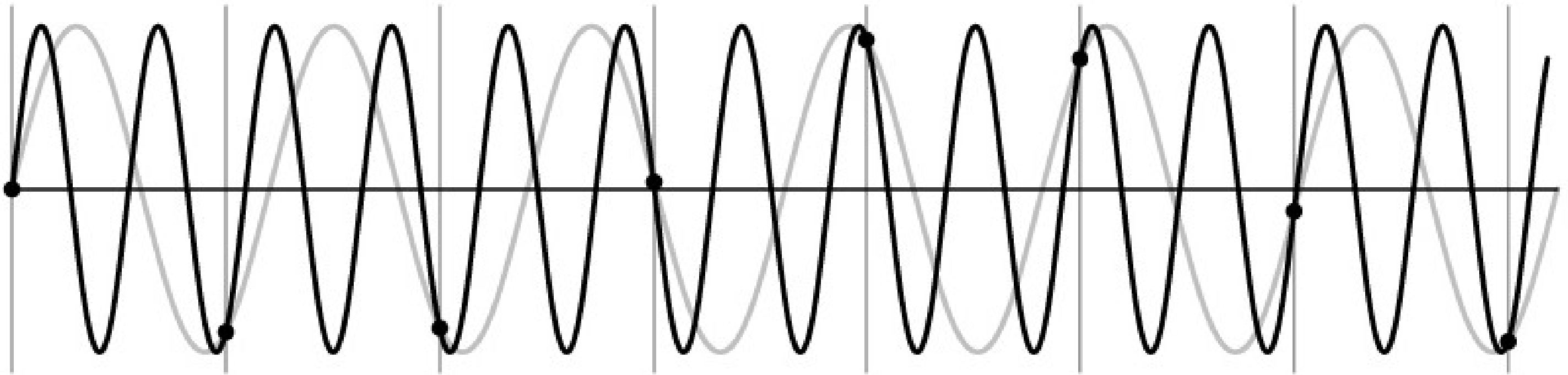


Unsurprising effect: **information is lost.**

Surprising effect: we can confuse the signal with one of **lower frequency.**

Undersampling

The same samples can also match a signal of higher frequency.

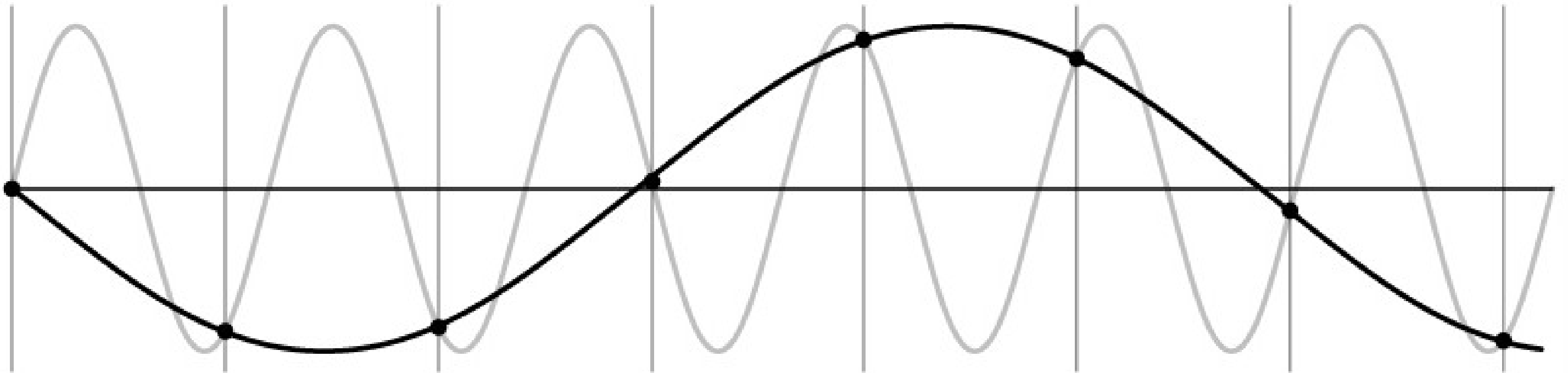


The samples alone do not uniquely identify the original signal.

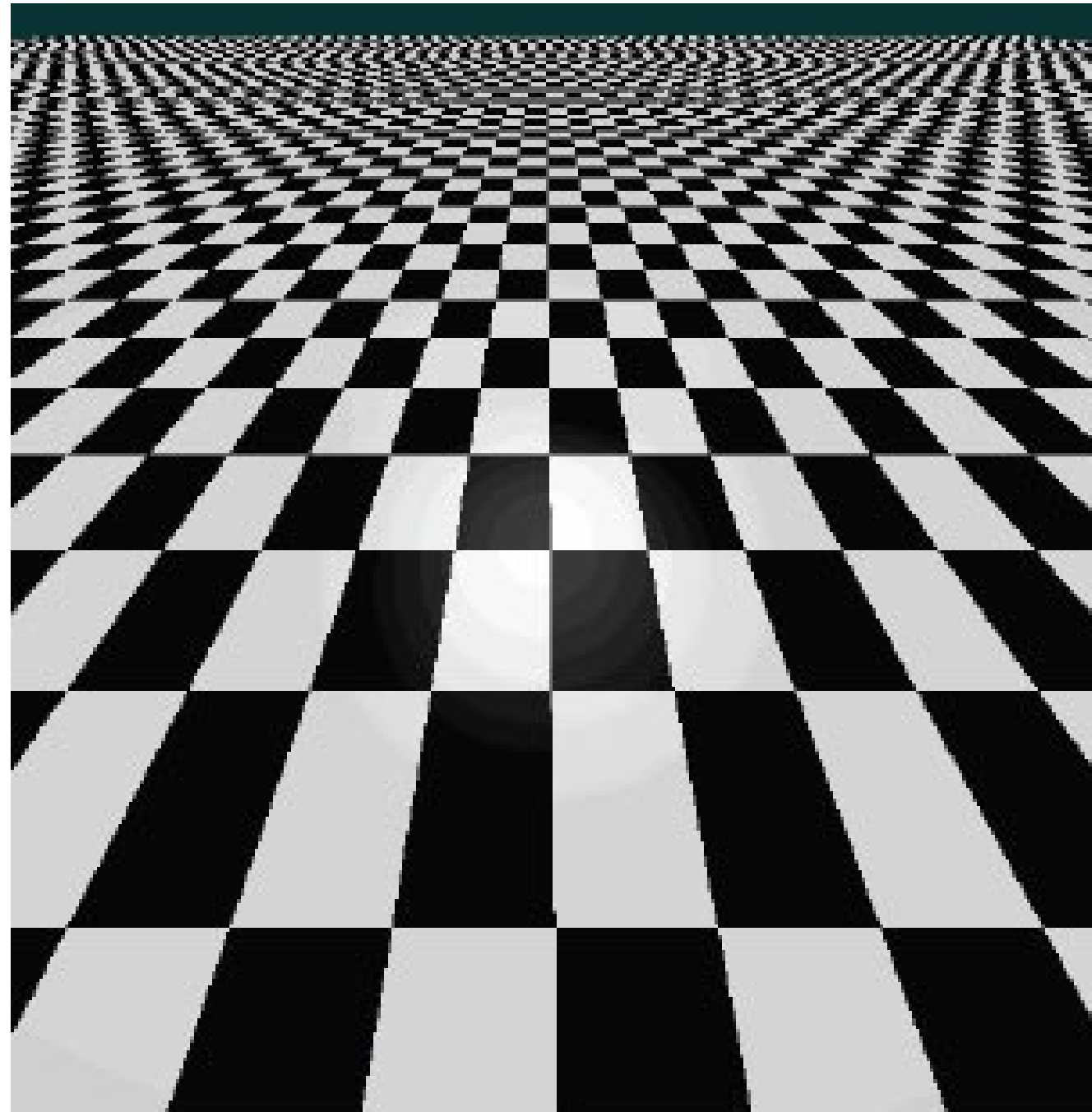
We need an assumption about its **highest frequency**.

Aliasing

Aliasing: Undersampling disguises a signal as one of a lower frequency.



Aliasing in Textures



Aliasing in Photographs

These patterns are also known as **moiré patterns**.



Striped fabric



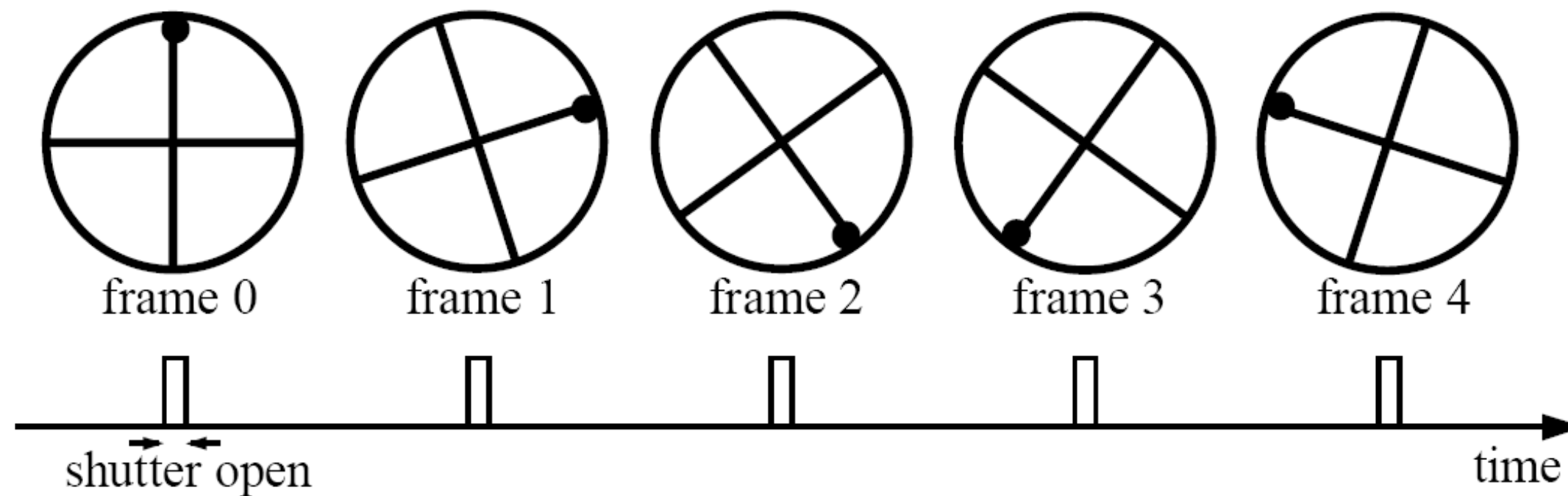
Building façades



AC grille

Temporal Aliasing

- The camera records **separate snapshots**; we do not see the motion between frames.
- Follow the **dot**: the wheel is rotating **clockwise**



- Without the dot, the four spokes are **indistinguishable**.
- A large clockwise step produces the **same spoke positions** as a small counterclockwise step.

The wheel can therefore appear to rotate backward—an example of temporal aliasing.

Temporal Aliasing: Wagon-Wheel Effect

 The wagon wheel effect
ns194



Watch on

Temporal Aliasing: Helicopter Rotor



camera shutter speed and frame rate match helicopter`s rotor
Chris Fay



Watch on

Anti-aliasing: Oversampling

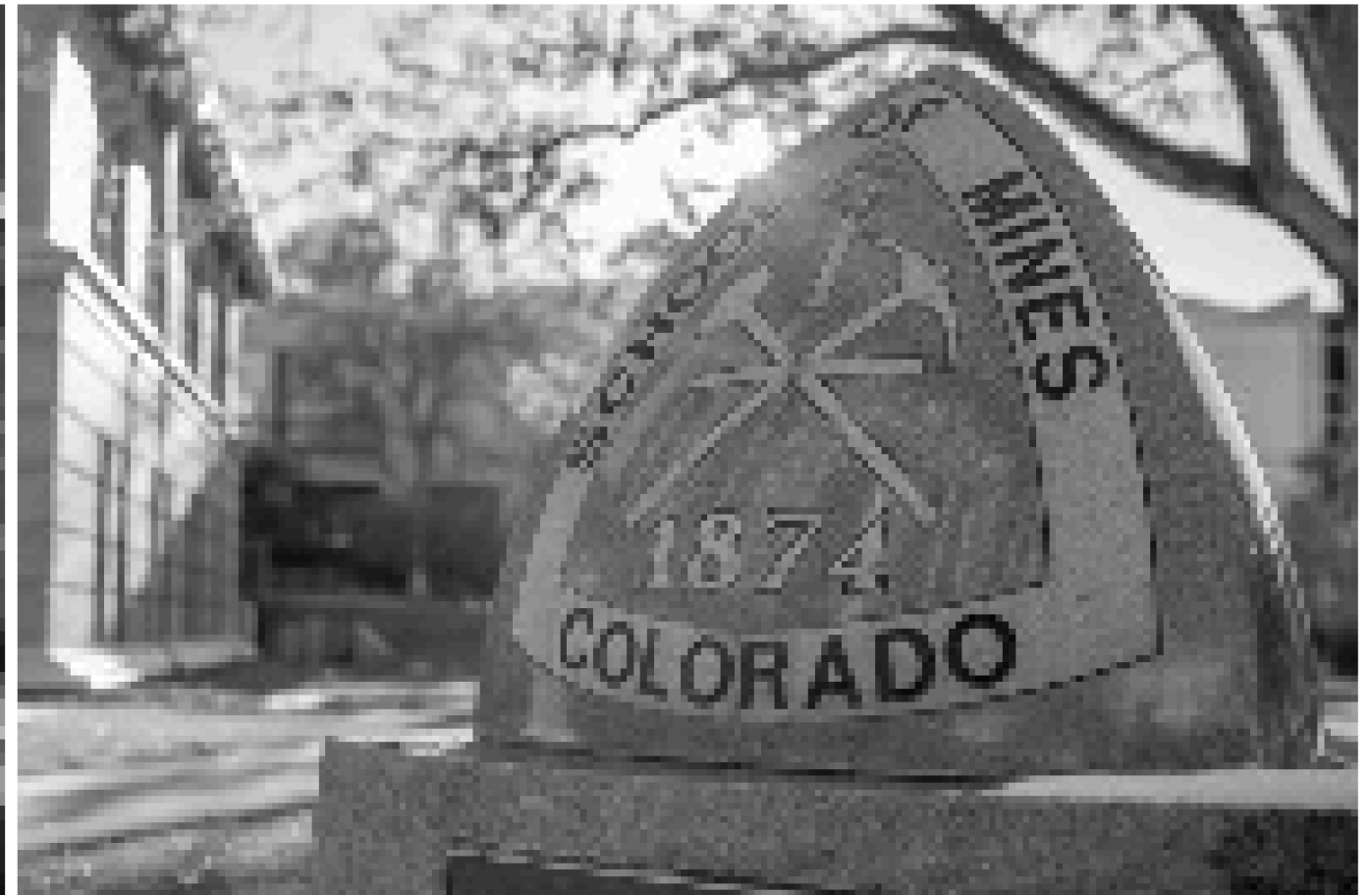
How would you deal with aliasing?

Approach 1: Use more samples to represent fine detail.

Naïve sampling: 1/4
100 × 67 pixels



More samples: 1/2
200 × 133 pixels



Anti-aliasing: Smoothing

Approach 2: Smooth the image **before** downsampling.

Direct downsampling: 1/4
100 × 67 pixels



Smooth, then downsample: 1/4
100 × 67 pixels



Some fine detail is lost, but aliasing artifacts are reduced.

How would you smooth a signal?

Better Image Downsampling

Apply a Gaussian filter, then keep every other row and column. Repeat.

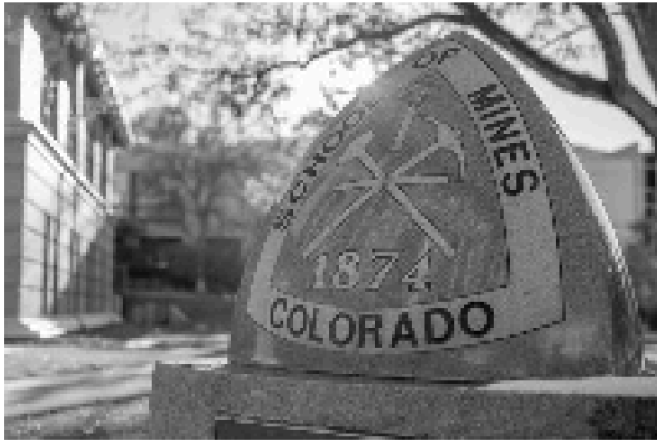
```
1 import cv2
2 import matplotlib.pyplot as plt
3
4 y2g = cv2.GaussianBlur(I, (3,3), 2)[::2, ::2]
5 y4g = cv2.GaussianBlur(y2g, (3,3), 2)[::2, ::2]
6 y8g = cv2.GaussianBlur(y4g, (3,3), 2)[::2, ::2]
7
8 fig, ax = plt.subplots(1, 3, figsize=(8, 4),
9     gridspec_kw={"width_ratios": [4, 2, 1]})
10 for a, y, s in zip(ax, [y2g, y4g, y8g], [2, 4, 8]):
11     a.imshow(y, cmap="gray", vmin=0, vmax=255,
12         interpolation="nearest")
13     a.set_title(f"1/{s}")
14     a.axis("off")
15 plt.tight_layout()
16 plt.show()
17 plt.close(fig)
```



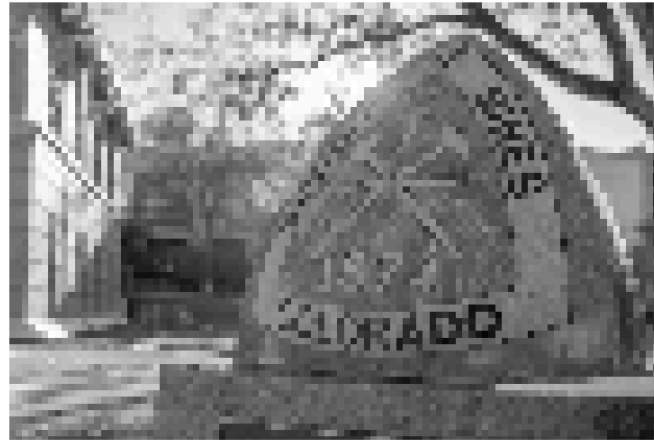
Comparison of Downsampling Methods

Naïve Image Downsampling:

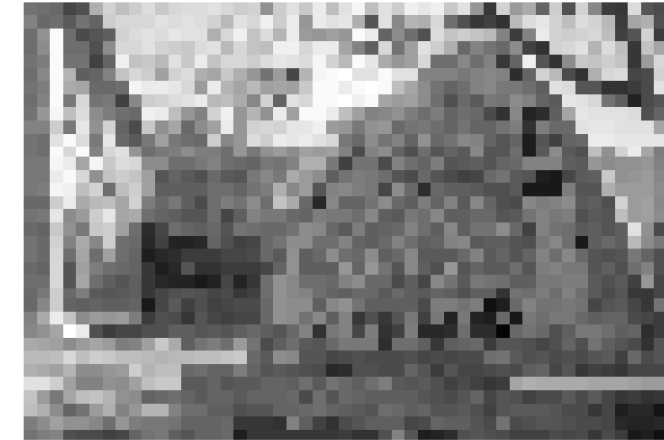
1/2



1/4 (2× zoom)

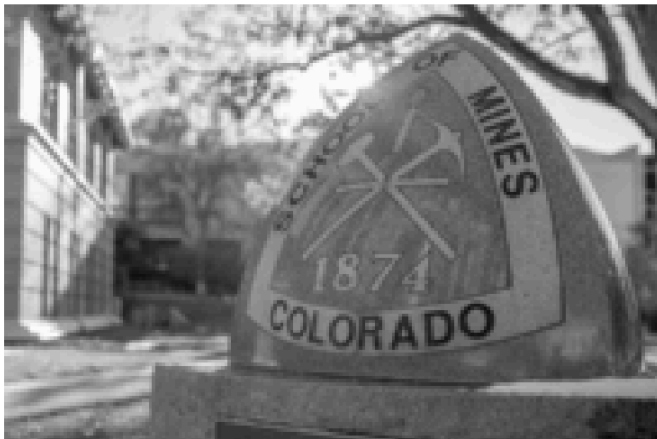


1/8 (4× zoom)



Smoothed Image Downsampling:

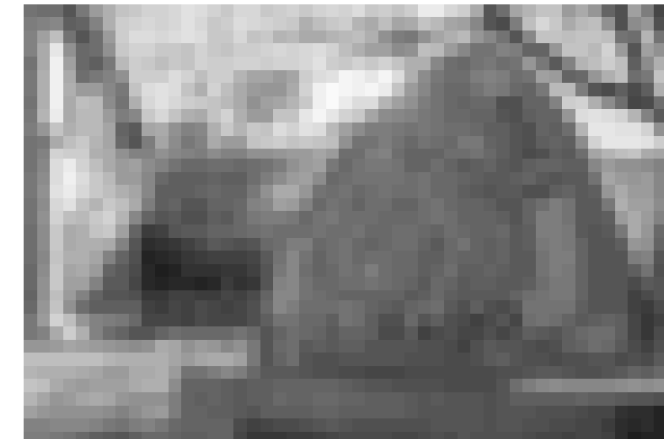
1/2



1/4 (2× zoom)



1/8 (4× zoom)



Anti-aliasing: How Much Is Enough?

- How much smoothing is enough?
- How many samples are enough?



Tip

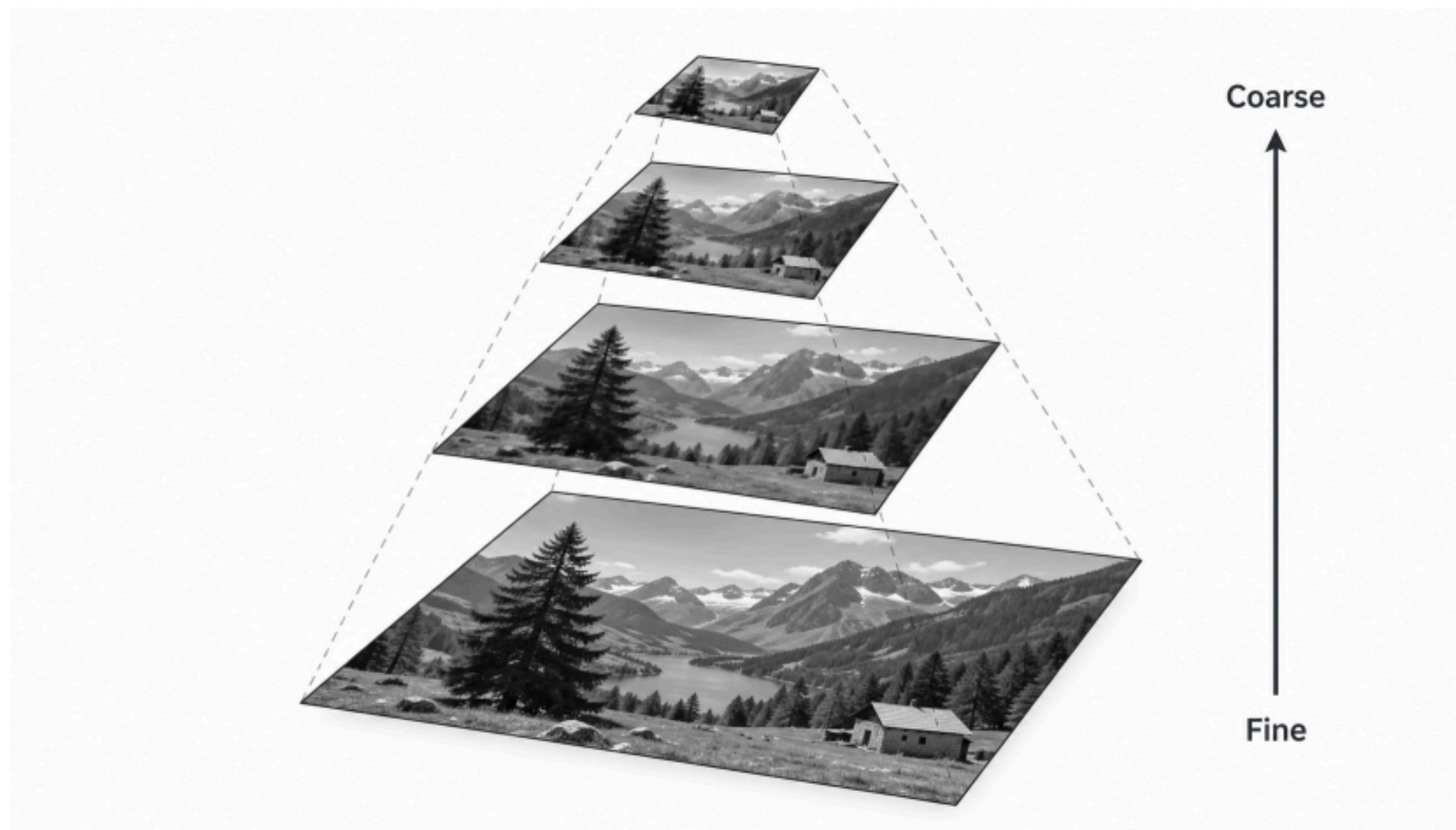
Both depend on the **Nyquist condition**:

$$f_s > 2f_{\max}$$

- f_s : sampling rate.
- $f_{\max}$: highest frequency.

We'll see what this means soon.

Image Pyramids



Why Use Image Pyramids?

Image compression



Texture mapping

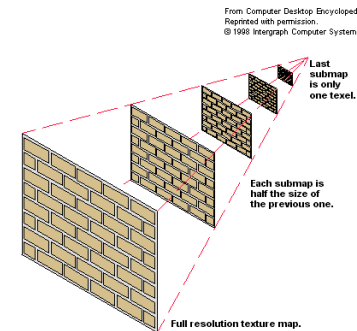


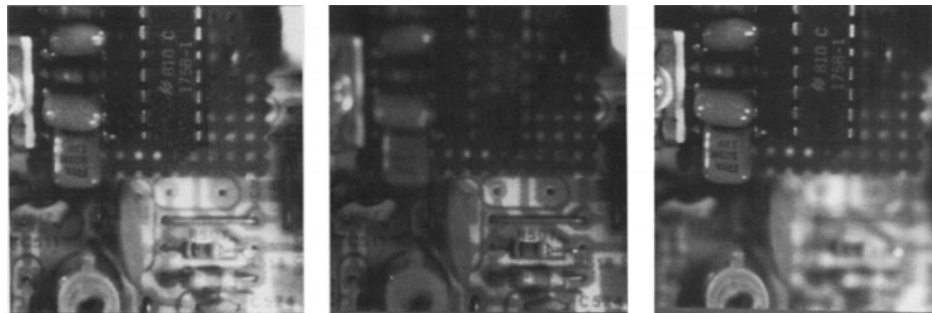
Image blending



Denoising



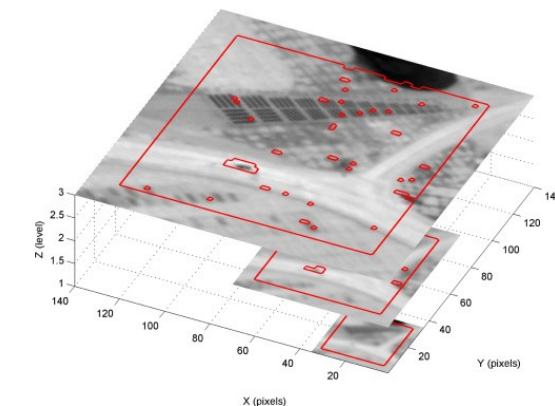
Focal stack compositing



Multiscale detection



Multiscale registration



Also: Optical flow and feature tracking estimate motion from a coarse to fine paradigm.

Gaussian Pyramid

A **Gaussian pyramid** represents an image at progressively coarser spatial scales.

$$G_0 = I, \quad G_{i+1} = \text{downsample}(h_G * G_i)$$

Construction

1. **Smooth** the current image using a Gaussian kernel h_G .
2. **Downsample** by a factor of 2, e.g., keep every other row and column.
3. **Repeat** until the desired minimum resolution is reached.

How many pixels does G_{i+1} have compared with G_i ?

Approximately $\frac{1}{4}$ as many pixels.

Why smooth first?

To reduce aliasing before downsampling.

Must the factor be 2?

No! Other scale factors are possible. A factor of 2 is conventional and is used by `cv2.pyrDown()`.

Constructing a Gaussian Pyramid

`cv2.pyrDown()` combines Gaussian smoothing and downsampling.

```
1 import cv2
2 import matplotlib.pyplot as plt
3
4 G = [I]
5 for _ in range(3):
6     y = cv2.pyrDown(G[-1], borderType=cv2.BORDER_REPLICATE)
7     G.append(y)
8
9 heights = [g.shape[0] for g in G[::-1]]
10 fig, ax = plt.subplots(4, 1, figsize=(10, 7.5))
11 gridspec_kw={"height_ratios": heights}
12
13 W = I.shape[1]
14 for a, i in zip(ax, range(3, -1, -1)):
15     h, w = G[i].shape
16     a.imshow(G[i], cmap="gray", vmin=0, vmax=255,
17              interpolation="nearest",
18              extent=((W-w)/2, (W+w)/2, h, 0))
19     a.set_xlim(0, W)
20     a.text(1.02, .5, rf"$G_{i}$", fontsize=20)
21     a.axis("off")
```



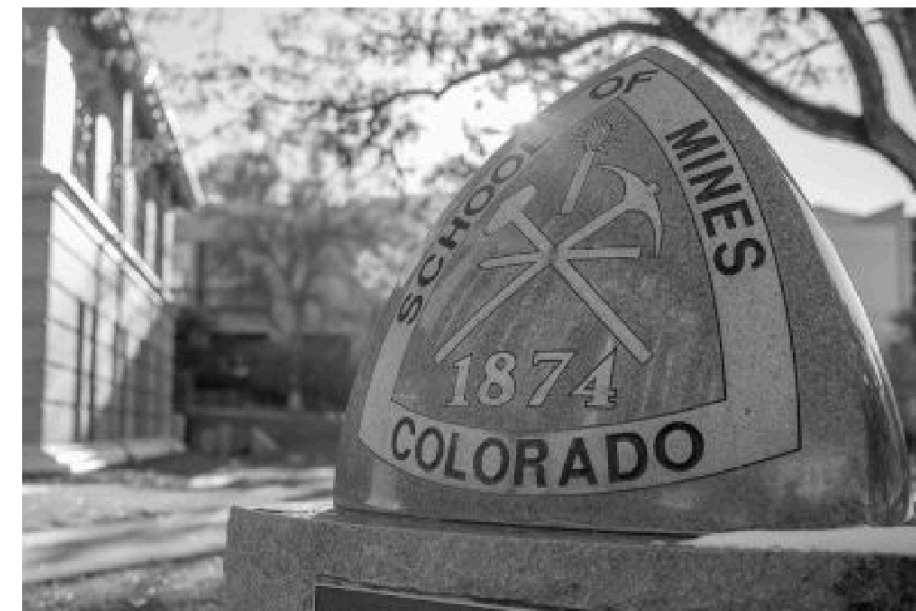
G_3



G_2



G_1



G_0

What Happens at Coarser Levels?

What happens to the details of the image?

- Fine details get **smoothed out** as we move to higher, coarser levels.
- Small structures and texture become less distinguishable.

What is preserved at the higher levels?

- Mostly **large uniform regions** and broad intensity variations.
- The overall shape and arrangement of large structures.



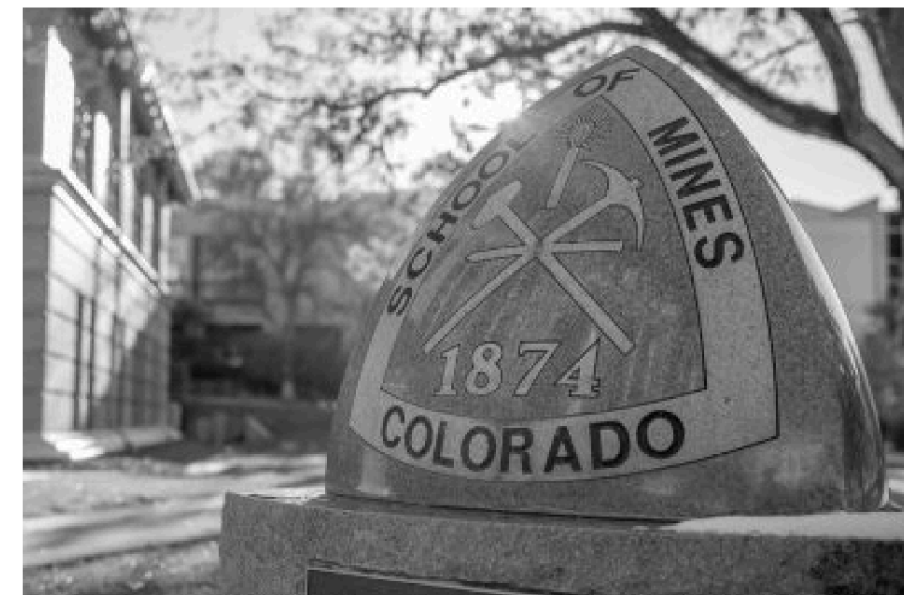
G_3



G_2



G_1



G_0

Can We Recover the Original Image?

How would you reconstruct the original image from an upper level image?

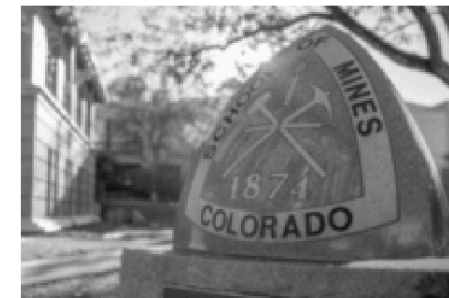
- Exact reconstruction is not possible!
- Smoothing suppresses fine details.
- Downsampling retains fewer samples.
- Enlarging the coarse image cannot recover the missing information!



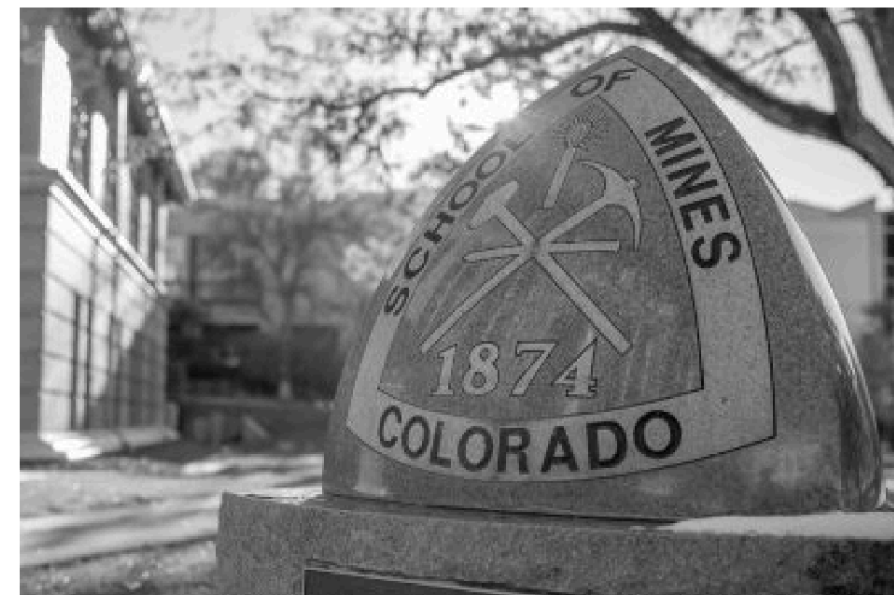
G_3



G_2



G_1



G_0

Blurring Removes Fine Detail

Gaussian smoothing suppresses detail **before we even downsample**.

Original — level 0



Smoothed — before downsampling



What does the residual look like?

The residual is the original image **minus** the smoothed image.

What Did Smoothing Remove?



- The residual contains **edges and fine texture** suppressed by smoothing.
- Mid-gray represents zero; brighter and darker values represent positive and negative differences.

Can we make a pyramid that allows exact reconstruction?

Yes! Retain the residual information along with the coarse image. This is the idea behind a **Laplacian pyramid**.

Start with One Level

The original image equals an **expanded coarse image** plus a **residual**.



Do we need to store every blurred image as well as every residual?

No. We can recover each finer image from its residual and the next coarser image.

Laplacian Pyramid

Construction

Start with $G_0 = I$. At each level:

1. **Smooth and downsample** to obtain G_{i+1} .
2. **Expand** G_{i+1} back to the size of G_i .
3. **Subtract** to obtain the residual:

$$L_i = G_i - \text{expand}(G_{i+1})$$

Repeat until the desired minimum resolution is reached.

Here, **expand** means upsample and filter, using `cv2.pyrUp()`.

- The sequence $G_0, G_1, G_2, \dots, G_n$ is the **Gaussian pyramid**.
- The sequence $L_0, L_1, \dots, L_{n-1}, G_n$ is the **Laplacian pyramid**.
- The Laplacian pyramid stores the detail needed to move back from one Gaussian level to the next finer level.

Upsampling: A Numerical Example

In Laplacian pyramid, **expand** means:

1. Insert zeros between rows and columns to enlarge the pixel grid.
2. Apply a Gaussian filter to interpolate pixel values.

Original image: A 2×2 image.

$$\begin{bmatrix} 32 & 96 \\ 160 & 224 \end{bmatrix}$$

Insert zeros: Double width & height.

$$\begin{bmatrix} 32 & 0 & 96 & 0 \\ 0 & 0 & 0 & 0 \\ 160 & 0 & 224 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Interpolation filter: `cv2.pyrUp( )`.

$$\begin{bmatrix} 80 & 96 & 120 & 128 \\ 112 & 128 & 152 & 160 \\ 160 & 176 & 200 & 208 \\ 176 & 192 & 216 & 224 \end{bmatrix}$$

- The filter spreads pixel values into neighboring positions.
- Its weights are scaled to compensate for the inserted zeros.

Note

Filtering changes values at the original pixel positions too. The enlarged image is a smooth estimate of the finer image.

Laplacian Pyramid in OpenCV

Use floating-point arrays to preserve negative residual values.

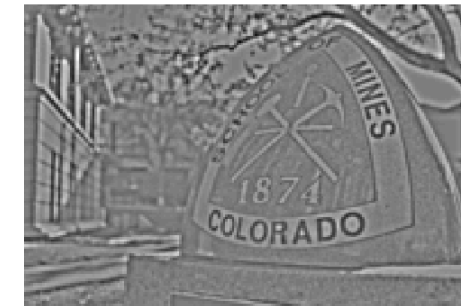
```
1 G = [I.astype("float32")]
2 L = []
3
4 for _ in range(3):
5     y = cv2.pyrDown(G[-1], borderType=cv2.BORDER_REPLICATE)
6     u = cv2.pyrUp(y, dstsize=G[-1].shape[::-1])
7     L.append(G[-1] - u)
8     G.append(y)
9
10 L.append(G[-1]) # Keep the smallest image to save space
11
12 heights = [im.shape[0] for im in L[::-1]]
13 fig, ax = plt.subplots(4, 1, figsize=(10, 7.5))
14         gridspec_kw={"height_ratios": heights})
15
16 W = I.shape[1]
17 for a, i in zip(ax, range(3, -1, -1)):
18     h, w = L[i].shape
19     s = max(float(abs(L[i]).max()), 1)
20     lo, hi = (0, 255) if i == 3 else (-s, s)
21     a.imshow(L[i], cmap="gray", vmin=lo, vmax=hi)
```



$L_3 = G_3$



L_2



L_1



L_0

What Must We Store?

What do we need to reconstruct the original image?

1. The residuals

$$L_0, \quad L_1, \quad L_2$$

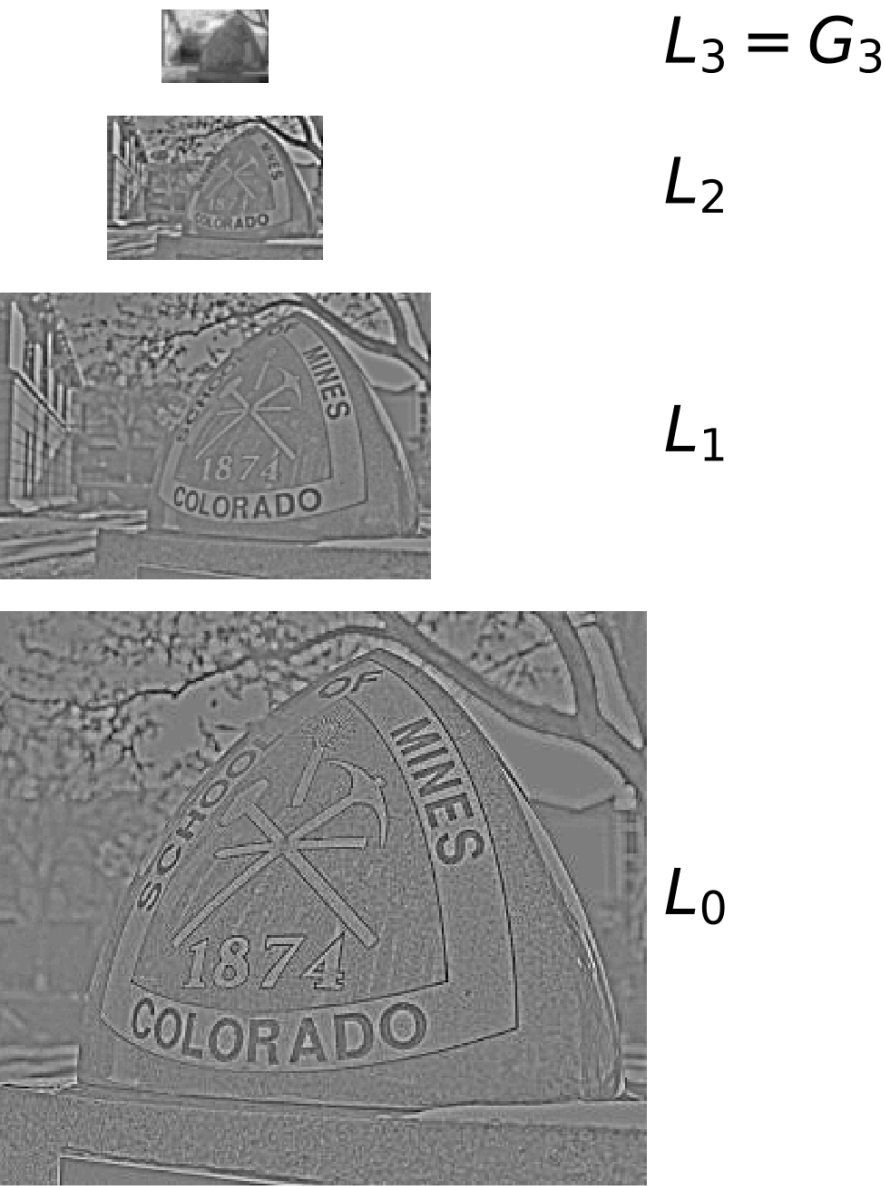
These retain the detail lost between consecutive Gaussian levels.

2. The smallest Gaussian image

$$L_3 = G_3$$

This provides the starting point for reconstruction.

The intermediate Gaussian images can also be recovered as needed.



Residual contrast is scaled for display; mid-gray means zero.
Reconstruction uses the original signed arrays.

Reconstructing the Original Image

Start with the smallest image.

At each step:

1. **Expand** to the next finer resolution.
2. **Add** the residual at that level.
3. **Repeat** until reaching the original resolution.

$$G_i = \text{expand}(G_{i+1}) + L_i$$

The residual restores the detail missing from the expanded image.

Keeping the residuals and smallest image allows **exact reconstruction**, apart from numerical roundoff.

Expand G_3



+

L_2



=

Recovered G_2



Expand G_2



+

L_1

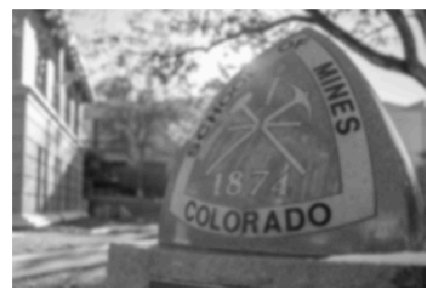


=

Recovered G_1



Expand G_1



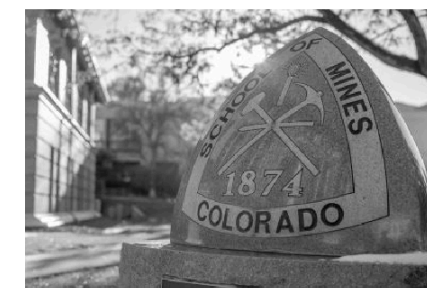
+

L_0



=

Recovered G_0



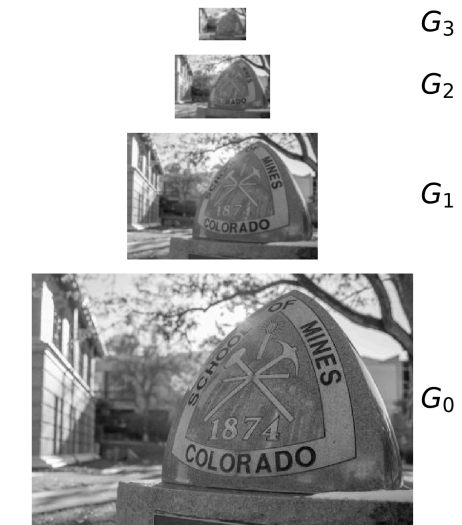
Gaussian vs. Laplacian Pyramid

Which pyramid takes more space to store?

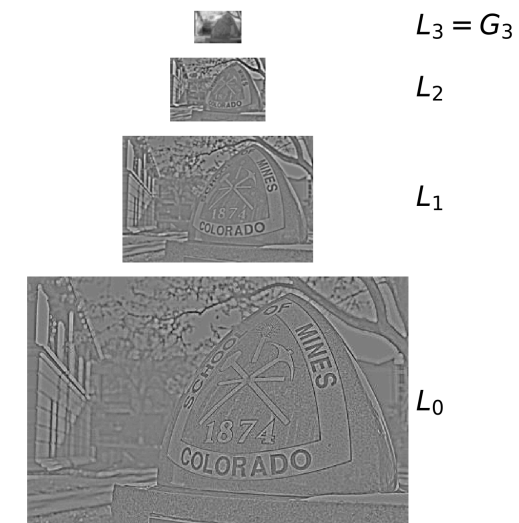
- Their raw storage is equal when using the same data type.
- A Laplacian pyramid can require **more bits per value** to represent signed differences and preserve precision.
- They take about **33% more than the original image**:

$$N + \frac{N}{4} + \frac{N}{16} + \dots \approx \frac{4}{3}N$$

Gaussian pyramid



Laplacian pyramid



Laplacian Pyramids Useful for Compression

Raw array size and compressed file size are different.

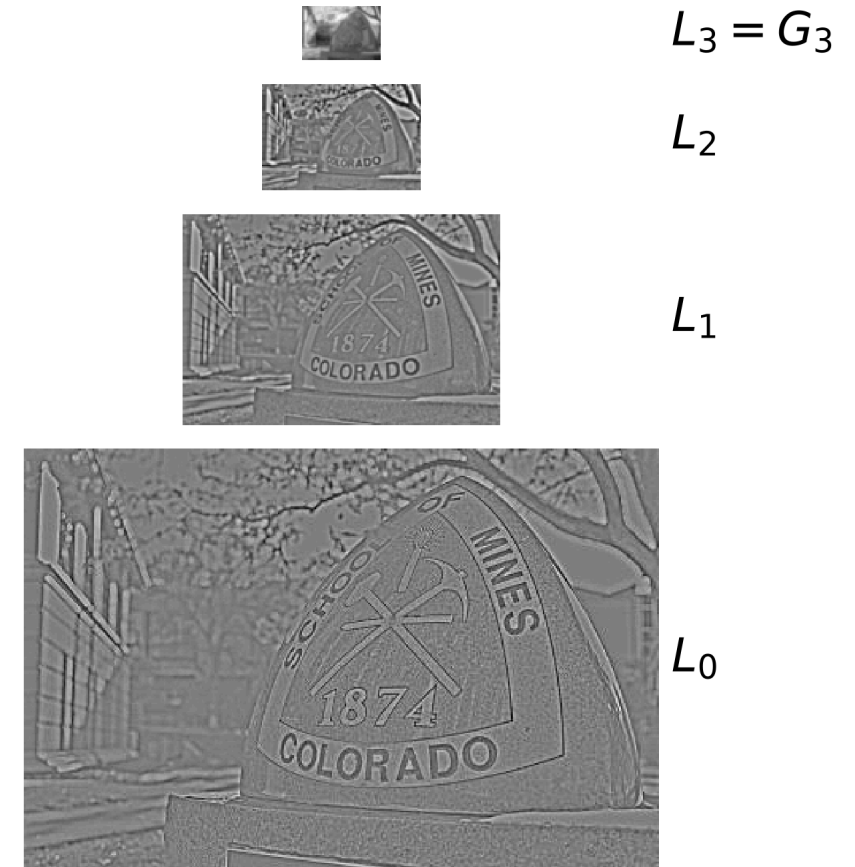
- Laplacian levels store **differences between scales**.
- In smooth regions, these differences are often **zero or close to zero**.

Frequently occurring values can be encoded using fewer bits

- The residuals can be inexpensive to store after compression, even though the raw pyramid contains more values than the original image.

Lossy compression

- We can round small residuals to zero or store them with less precision.
- This reduces storage further, at the cost of some image detail.
- Actual savings depend on the image and the encoding method.



Where Are Image Pyramids Used?

Application	How pyramids help
Panorama stitching and blending	Combine images at multiple scales to reduce visible seams.
Exposure fusion	Combine well-exposed regions from photographs taken at different exposures.
Motion estimation and feature tracking	Estimate motion at coarse resolutions, then refine it at finer resolutions.
Detail and local-contrast adjustment	Strengthen or suppress image details at selected spatial scales.

Application: Selective Detail Enhancement

Original



Foreground mask



Edited result



- Amplify foreground residuals, attenuate background residuals, then reconstruct:

$$\tilde{L}_i = [\alpha M_i + \beta(1 - M_i)] L_i, \quad \alpha = 1.5, \quad \beta = 0.25$$

- M_i is the foreground mask at level i : white selects the foreground, black selects the background, and gray gives a gradual transition.

Steerable Pyramids

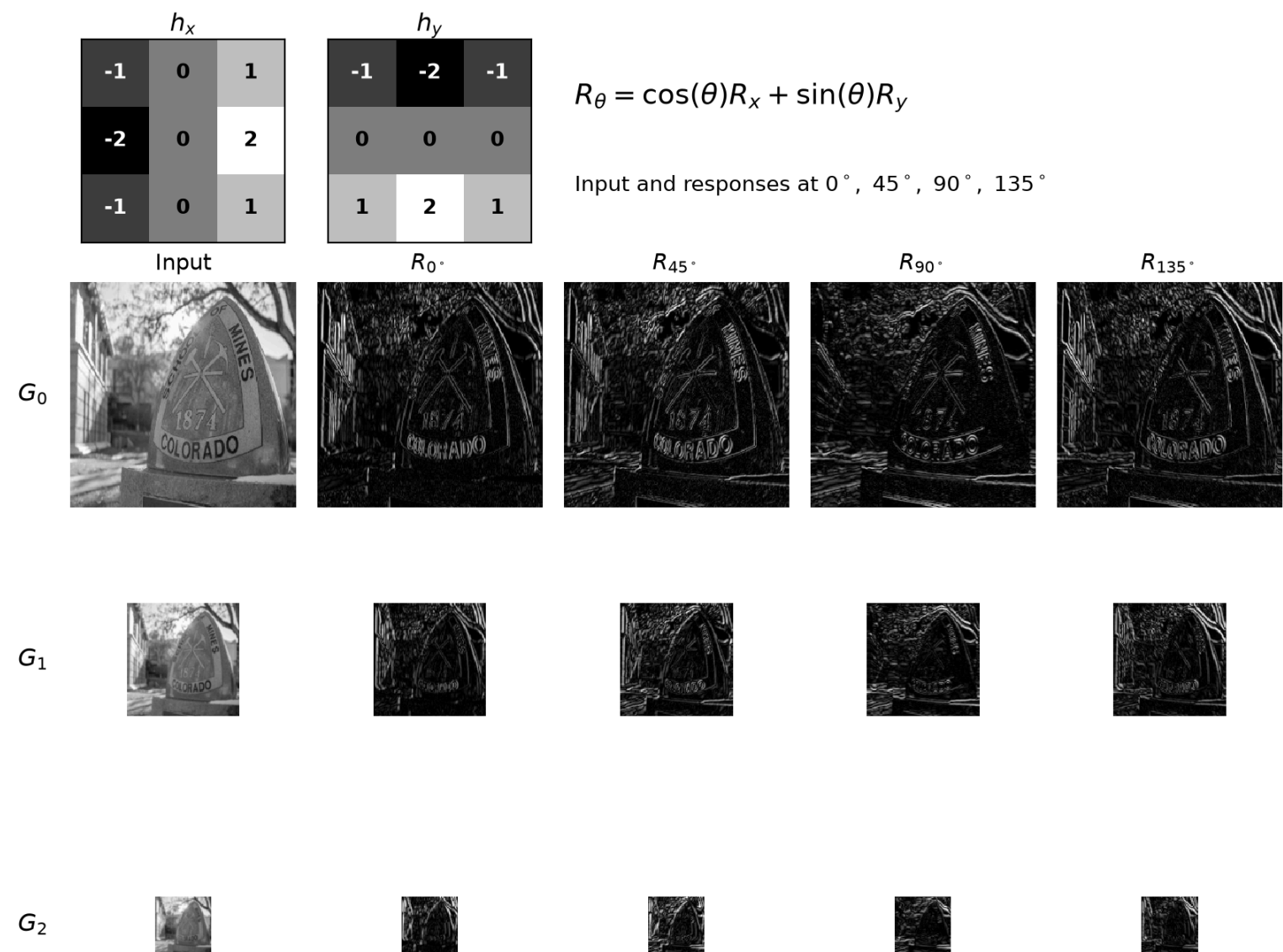
- A multi-scale, multi-orientation image representation.
- Each pyramid level contains responses to structures at several orientations.
- A filter is **steerable** when the response at an arbitrary orientation can be synthesized from a small set of basis filters.

For first-order derivative **basis responses**:

$$R_{\theta} = \cos(\theta) R_x + \sin(\theta) R_y$$

Applications

- Edge and contour analysis
- Texture and orientation analysis
- Image denoising and enhancement
- Image fusion
- Computational models of visual perception



A simplified **first-order steerable pyramid**: rows correspond to Gaussian pyramid levels G_0 , G_1 , and G_2 , while columns show orientation-selective responses synthesized from the basis responses R_x and R_y .

Wavelets

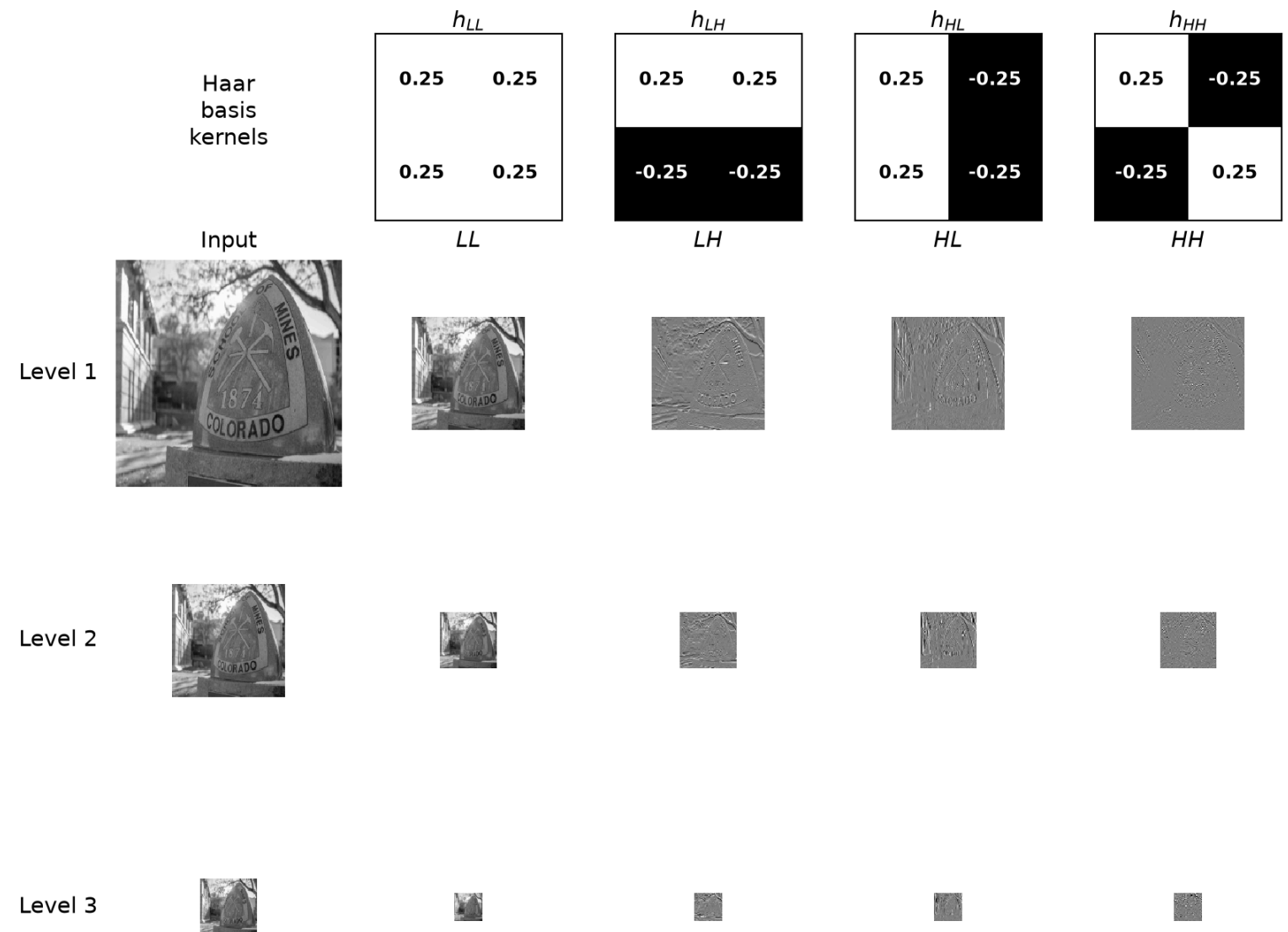
A **wavelet transform** separates an image into spatial-frequency subbands:

- **LL**: coarse approximation
- **LH**: horizontal detail
- **HL**: vertical detail
- **HH**: diagonal detail

The **LL** image can be decomposed again, producing a multi-scale representation.

Applications

- Image compression — e.g., **JPEG 2000**
- Image denoising
- Feature extraction
- Image fusion
- Medical and scientific imaging



Haar wavelet decomposition: at each level, the input is separated into a low-frequency approximation LL and three directional detail bands. The LL band becomes the input to the next level.

Key Takeaways

- Downsampling reduces image size, but naïve subsampling can discard important structure and create **aliasing**.
- Anti-aliasing smooths the image before downsampling so high-frequency detail does not fold into lower frequencies.
- A **Gaussian pyramid** stores progressively smaller, smoother versions of an image.
- A **Laplacian pyramid** stores what is lost between Gaussian pyramid levels: the image details or residuals.
- Laplacian pyramids can reconstruct the original image when the residuals and coarsest image are preserved.
- Multi-scale image representations are widely used in compression, blending, tracking, enhancement, steerable pyramids, and wavelets.